

NoTeX

A modern noteworthy LaTeX template

Multi File Example

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1 Introduction



Welcome to NoTeX [1], a modern $\text{\LaTeX}$ template designed to make your note-taking and document creation experience enjoyable and efficient. Whether you're writing notes for your next project, the documentation for your code, or just enjoy the look of this template, NoTeX has you covered.

The idea behind this template is to take care of all the formatting, aesthetics, and structure for you, so you can focus on the content itself and organize your notes in a clean and visually appealing way. NoTeX comes with a set of pre-defined styles, useful commands and commonly used macros to help you create professional-looking documents with minimal effort.

The template is designed to be modular, allowing you to easily include or exclude components as needed, and customizable, so you can adapt it to your own style and preferences.

Aside from the LaTeX template itself, NoTeX also provides a fast *command line interface* (CLI) companion utility written in C++ to manage template installation, project initialization, document compilation, and more... This can be optionally installed on your system to streamline your workflow and make it even easier to use NoTeX.

The following sections in this document will guide you through the usage of the framework, from its installation to the creation and management of your own projects with NoTeX, showcasing the various features and capabilities of this template.

If you like this template please visit my repository on GitHub [1] and give it a star ★.

 [marcotallone/notex](https://github.com/marcotallone/notex)

Happy writing!

2 Installation

This section will guide you through the installation process of NoTeX and its companion utility.

2.1 Requirements

The project requires the following dependencies installed in your system:

- CMake (≥ 3.20)
- C++ compiler (C++20)
- LaTeX distribution, with the following packages:
 - `texlive-latex-base`
 - `texlive-latex-extra`
 - `texlive-fonts-recommended`
 - `texlive-fonts-extra`
 - `texlive-luatex`
 - `texlive-lang-english`
 - `texlive-bibtex-extra`
 - `texlive-science`
 - `biber`
 - `latexmk`
- LuaLaTeX compiler: this can be installed with the above `texlive-luatex` package
- Pygments python package: can be installed with `pip install Pygments` in your preferred environment

2.2 CLI Utility Installation

2.2.1 Building from source

The NoTeX software can be built from source following these steps:

1. Clone the repository:

```
git clone https://github.com/marcotallone/notex.git
```

2. Go to the project's root:

```
cd notex
```

3. Compile using cmake:

```
mkdir -p build \  
&& cmake -S . -B build -DCMAKE_BUILD_TYPE=Release \  
&& cmake --build build --parallel 4
```

This will produce a unique binary in `./bin/notex`, containing the CLI utility with embedded LaTeX template.

4. *Optionally*, you can install the software system-wise for easier usage:

```
sudo cmake --install build
```

This will allow to use `notex` as a normal binary everywhere in your system.

 The system-wise installation might require `sudo` privileges!

5. Check if your installation was successful by running:

```
notex info
# or, if you skipped step 3:
# ./bin/notex info
```

The `[install.sh](../install.sh)` script provided in the project's repository helps to automate these steps with:

```
./install.sh # for local compilation only
```

or:

```
sudo ./install.sh global # for system-wise installation
```

 Mind that the latter option also installs the LaTeX template system-wise, see below for further details.

2.2.2 Build Options

When building from source, the CMake compilation offers multiple options that are not set by default:

- `DEBUG`: enables debug mode, with extended logging reporting.
- `VERBOSE`: enables verbose mode.
- `LOGGING`: enables logging reporting. Logs are always stored in the `./log/notex.log` file in the project where `notex` is used.
- `TESTS`: enables tests. See `[Testing](../Testing.md)` for details.
- `COVERAGE`: enables coverage report. See `[Testing](../Testing.md)` for details.
- `DOCS`: allows to build project documentation providing the `docs` target. After setting this option with `-DDOCS=ON`, documentation can be built in `./build/docs/` with:

```
cmake --build build --target docs
```

2.2.3 Uninstall

You can uninstall a global `notex` installation with the command:

```
notex uninstall
```

Or, equivalently:

```
notex prune
```

A local installation can be analogously uninstalled by providing the path to the project installation:

```
notex uninstall <path>
```

2.3 LaTeX Template Installation

After installing the `notex` CLI utility, the LaTeX template can be installed either **globally** (system-wise available) or **locally** (for the single LaTeX project).

2.3.1 Global Installation

The template can be globally installed with:

```
notex install
```

Which will install the LaTeX classes and styles, as well as required fonts, into your TeX distribution's `TEXMFHOME` directory. To check which directory this is in your own system, you can run:

```
kpsewhich -var-value=TEXMFHOME
```

 Pass `--force` to overwrite an existing installation without being asked to confirm.

After a global installation the template can be loaded as:

```
\documentclass[OPTIONS]{notex}
```

2.3.2 Local Installation

You can install the template locally, for your project only, by running:

```
notex install <path>
```

where `<path>` is the path to your LaTeX project root directory. For instance if the root directory of your project is the current one run:

```
notex install .
```

The local installation will install the template and its classes in the `<path>/settings/` directory, while fonts will be placed in `<path>/fonts/`.

After a local installation the template can be loaded as:

```
\documentclass[OPTIONS]{settings/notex}
```

3 Usage

Once the `notex` CLI utility is installed, it offers a set of commands to facilitate managing and working with your $\text{\LaTeX}$ projects. The following table summarizes the main commands and their purpose. Further details can be obtained by running `notex help` (or `notex -h`).

Command	Description
<code>notex help</code>	Shows <code>notex</code> binary options and help guide.
<code>notex version</code>	Shows current binary version.
<code>notex info</code>	Displays installation and current project information.
<code>notex init .</code>	Initializes a project in the current working directory.
<code>notex clean <path></code>	Cleans recursively temporary files from the provided <code><path></code> .
<code>notex get <snippet></code>	Provides a ready-to-use $\text{\LaTeX}$ snippet template.
<code>notex theme <theme></code>	Changes the current project theme.
<code>notex add <what></code>	Adds a new section or bibliography to the current project.
<code>notex remove <what></code>	Removes a section or bibliography from the current project.
<code>notex ls</code>	Lists the files that belong to the current project.
<code>notex checkhealth</code>	Checks the $\text{\TeX}$ environment, installation, and project state.
<code>notex set <key> <what></code>	Changes a whitelisted <code>notex.json</code> key directly.
<code>notex reset</code>	Regenerates <code>main.tex</code> from the template.
<code>notex delete</code>	Removes notex files and build artefacts, keeps your documents.

 This guide assumes that the `notex` utility has been installed as a globally available binary.

3.1 Initialization

You can quickly set up a $\text{\LaTeX}$ project in the current directory (*or in one of its sub-paths*) via the **init command**:

```
notex init .  
# or equivalent:  
# notex init multi .
```

The default initialization starts a **multi-file** project: a `main.tex` using the `subfiles` package is created together with a `./sections/` folder for individual section files. An example of this layout can be found in the `examples/multi/` directory. For smaller projects, a **mono-file** project with a single `main.tex` can be initialized instead:

```
notex init mono .
```

An example can be found in the `examples/mono/` directory. Both commands also create a `.notex/` directory at the project root containing NoTeX configuration.

 `notex init` refuses to overwrite an existing `main.tex` or `.notex/` directory. Pass `--force` to overwrite anyway.

In both cases, the generated `main.tex` automatically uses `\documentclass{settings/notex}` when a local installation is detected in the target directory, and `\documentclass{notex}` otherwise.

3.2 Management

Once you have a working project, `notex` provides a series of management commands to facilitate common operations.

3.2.1 Change Theme

You can change the theme used by the $\text{\LaTeX}$ template with:

```
notex theme <theme_name>
```

where `<theme_name>` can be one of the available themes:

- `light` (default)
- `dark`
- `tokyo`
- `bw`

3.2.2 Manage Sections and Bibliography

You can add a new section with:

```
notex add section "My Title"
```

This adds a section directly to `main.tex` for a mono-file project, or creates a new section file under `sections/` for a multi-file project.

You can remove a section by its number `<N>`:

```
notex remove section <N>
```

Two analogous commands exist for bibliography management.
To create `bibliography.bib` and wire it into `main.tex`:

```
notex add bib
```

To remove the bibliography:

```
notex remove bib
```

3.2.3 Cleaning

$\LaTeX$ projects accumulate temporary files and build artefacts. `notex` provides a command to clean them up recursively:

```
notex clean <path>
```

 Cleaned files are **lost forever** once the command runs. To preview which files would be removed **before actually deleting them**, run a *dry run*:

```
notex clean --dry-run
```

This lists the files that would be removed by an actual clean operation.

3.2.4 References and Snippets

To recall the $\LaTeX$ environments provided by the NoTeX template, the `get` command prints ready-to-paste snippet templates:

```
notex get
```

Lists every available snippet name (*info*, *warning*, *cbox*, *ebox*, *code*, ...). To fetch a specific snippet, pass its name:

```
notex get box      # displays a template for the "box" environment
notex get warning  # displays a template for the warning callout
notex get code     # displays a template for the code environment
```

3.3 State and Info

Several commands allow inspecting and modifying the current project state.

```
notex ls
```

Lists the files that belong to the project (*main file*, *sections*, *bibliography*).

```
notex checkhealth
```

Checks the $\TeX$ environment, the installation, and the project's state.

```
notex set main_file alt.tex
```


Changes a whitelisted `notex.json` key directly (e.g. `main_file`, `theme`, `bibliography_file`).

```
notex reset
```

Regenerates `main.tex` from the template, backing up the existing file to `main.tex.bak`.

```
notex delete
```

Removes `.notex/`, `settings/`, `fonts/`, and build artefacts while keeping your documents untouched. To remove the entire project scaffolding destructively:

```
notex delete --all
```

4 Callout Blocks

The NoTeX template provides a set of callout blocks that can be used to highlight important information, warnings, definitions, theorems, and more. These blocks are designed to improve the readability and organization of your document.

This section briefly shows how to use the various callout blocks available in the NoTeX template. You can look at the documentation source code that generated this file to see how each callout block is invoked.

4.0.1 Boxes

The most basic callout blocks are the boxes, which can be used to highlight text or separate content from the main body of the document. The following examples demonstrate different types of boxes.

Colored Box: A simple box with no additional formatting.

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This (and all the other callouts) also allow to have **upper and lower parts**, separated by a dashed line.

Upper section

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Lower section

Of course in colored boxes there is an option to change the background color easily.

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ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Empty Box: A box with empty background but colored border. Where the color of the border can of course be changed via the optional argument.

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Warnings: These can be used to state information that requires special attention. Warnings can be used to highlight potential issues, important notes, or any other information that the reader should be aware of.

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You can also change the background color of the warning box to make it more visually distinct.

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ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Info Boxes: Classical information boxes for notes or tips.

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For very important information, you can use the “[blueinfo box](#)”, which is a more visually distinct version of the info box.

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Note Boxes: A note box is identical to an info box. It has been implemented exclusively for ease of use and to provide a more intuitive name for the box.

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lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Tip Boxes: A useful callout to provide tips or suggestions to the reader. It can be used to highlight best practices, shortcuts, or any other helpful information.

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Important Boxes: An alternative way to highlight important information.

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Caution Boxes: These are similar to the warning boxes and simply provide an alternative different way to convey warnings and alert the reader to potential issues or important information.

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Example Boxes: These can be used to provide examples or illustrations of concepts discussed in the text, but that are less relevant in the main body of the document.

 EXAMPLE 4.1

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Math Blocks: For mathematical definitions, theorems, corollaries and propositions of various kind, the template implements dedicated blocks that clearly state the importance and the category of their content.

 def 4.1 Triangle

A triangle is a polygon with three edges and three vertices. Triangles are classified in:

- Equilateral: all sides are equal.
- Isosceles: two sides are equal.
- Scalene: no sides are equal.

 THEOREM 4.1 Pythagorean Theorem

In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides.

$$a^2 + b^2 = c^2$$

 COROLLARY 4.2 Converse of the Pythagorean Theorem

If the square of the length of one side of a triangle is equal to the sum of the squares of the lengths of the other two sides, then the triangle is a right triangle.

 PROPOSITION 4.3 Sum of angles in a triangle

The sum of the interior angles of a triangle is always equal to 180 degrees.

 LEMMA 4.4 Triangle Inequality

In any triangle, the length of any side must be less than the sum of the lengths of the other two sides.

5 Code Blocks

The NoTeX template provides a clean way to write code snippets in your documents. Everything is built on **minted**, which highlights code automatically through the **pygments** library, so you get proper syntax colouring for free in any of the languages **pygments** knows. The only exception is **pseudocode**, which **pygments** has no lexer for and which therefore gets an environment of its own.

5.1 Inline Code

There are two ways to write code inline. The `\cc` command gives you plain, markdown-style monospaced text with a coloured background and no highlighting at all: `print("Hello, World!")`. The `\inline` command instead runs the text through **minted**, so it comes out highlighted like any other block: `print("Hello, World!")`. The language goes in the optional argument.

5.2 Code Blocks

Every language has an environment named after it. Just open it and write, no arguments needed.

```
def hello(name):  
    """Greets the user by name."""  
    print(f"Hello, {name}!")
```

```
#include <iostream>  
void hello(const std::string& name) {  
    std::cout << "Hello, " << name << "!" << std::endl;  
}
```

If the language you need has no environment of its own, use the generic `code` environment and name the language yourself. Any name **pygments** recognises works.

```
fn hello(name: &str) {  
    println!("Hello, {name}!");  
}
```

Some code languages have different default settings than others. For instance, **YAML** blocks have tabs replaced with two spaces, following standard **YAML** style.

```
name: "Pippo"  
surname: "Pallo"  
age: 24  
list:  
  - item1  
  - item2  
  
# This is a comment  
nested:  
  key1: value1  
  key2: value2
```

Code 5.1: A **YAML** code block

5.3 Captions and References

Code blocks are captioned exactly like figures and tables. Add a `caption` key and the block gets a numbered caption underneath; add a `label` key too and you can point at it from anywhere in the text. Numbering follows the section, so the blocks below are 5.2 and 5.3, and it resets at every new section.

```
def hello(name):  
    """Greets the user by name."""  
    print(f"Hello, {name}!")
```

Code 5.2: A function greeting the user by name

A block without a caption is not numbered, just like a figure without one. Math mode works inside comments through the usual `$...$` delimiters.

```
def square(x):  
    """Returns the square of a number."""  
    return x**2 # returns  $x^2$ , the square of  $x$ 
```

Code 5.3: Squaring a number

Referencing works with every mechanism you would expect: code 5.2 and Code 5.3 both resolve, and `\listoflistings` prints a list of every captioned block in the document.

5.4 Pseudocode

Pseudocode gets its own environment. It runs on **listings** rather than **minted**, which buys the one thing pseudocode really needs and **minted** cannot offer: real math mode anywhere in the body, not only inside comments. Keywords common to pseudocode are automatically highlighted with some predefined words.

```
function shuffle(A):  
    // Fisher-Yates shuffle  
    N ← length(A)  
    for i = 0 to N-1 do  
        j ← random integer between 0 and N - 1  
        swap A[i] and A[j]  
    return A
```

Code 5.4: Shuffling a list in place

You can also write pseudocode with line numbers. By default, line numbers are hidden, but you can show them with the `linenos = true` key as in the following example:

```
1 if x > 0 then  
2   return  $\sqrt{x}$   
3 else  
4   error "negative input"
```

5.5 Options

All the code environments take the same optional keys, which you can mix freely.

Key	What it does
<code>caption</code>	Adds a numbered caption below the block
<code>label</code>	Makes the block referenceable with <code>\ref</code> or <code>\cref</code>
<code>linenos</code>	Shows line numbers (<code>linenos=false</code> hides them)
<code>options</code>	Extra raw minted options, or listings options for pseudocode
<code>tcb</code>	Extra raw tcolorbox options, to restyle the box itself

Table 5.1: The optional keys accepted by every code environment

As an example, consider the pseudocode language above with active line numbers. In that case, you could optionally make line numbers start at 10, increase left margin and change the box color to the background document color by passing the required keys to the `options` and `tcb` keys respectively:

```
10 if x > 0 then
11     return  $\sqrt{x}$ 
12 else
13     error "negative input"
```

Code 5.5: An example of custom code block with different options.

If you want any specific setting applied to every block in the document, set it once in the preamble with `\notexcodeset`, which takes the same keys. For instance `\notexcodeset{linenos}` numbers every code block you write.

 The opening bracket of the optional argument must sit on the same line as `\begin{...}`, with no space in between, and nothing else may follow the environment name on that line. This is what lets a block start with a character LaTeX would normally choke on, like the `#include` in the C++ example above.

References

- [1] Marco Tallone. Notex. URL: <https://github.com/marcotallone/notex>.